Chacha Chen

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Research Keywords

- ❏ **Applied Machine Learning**
- ❏ Reinforcement Learning
- ❏ Healthcare
- ❏ **Human Computer Interaction**
- ❏ Natural Language Processing
- ❏ Crowdsourcing

Education

Pennsylvania State University Aug. 2019 - Dec. 2020 (expected)
College of Information Sciences and Technologies
Master in Informatics

- **Research Topics**:
  - **Human Computer Interaction:** combine AI with crowdsourcing to create systems that are usable, robust, and intelligent.
  - **Applied Machine Learning**: explore topics related to applied machine learning in various fields including natural language processing, spatial-temporal data mining, healthcare and etc.
- **Advisor**: Ting-Hao (Kenneth) Huang

Shanghai Jiao Tong University Sep.2015 – Jun.2019
Department of Computer Science
Bachelor of Science in Computer Science

- **Advisor**: Weinan Zhang, Zhenhui (Jessie) Li

Technical University of Munich Apr.2017 – Sep.2017
Exchange Student in Department of Informatics

Publications

Preprints

1. Jiaqi Wang, **Chacha Chen**, Hua Shen, Frank E Ritter
   Toward an Interactive Online Course Framework
   To be submitted to CHI Case Studies 2021
2. Tongan Cai*, **Chacha Chen***, Ting-Hao Huang, Frank E Ritter
   Quantitative Analysis of Bibliography Management Applications
   To be submitted to CHI Case Studies 2021
3. **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li
   Physics-informed Generative Adversarial Networks
   Submitted to NeurIPS Workshop 2020, under review.
4. **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao
   A Bayesian Machine Learning Model for Healthcare
   Submitted to AAAI 2021, under review.
5. Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Zhenhui Li
   A Residual Model on Various Tasks
   Submitted to AAAI 2021, under review.

Conference Papers

1. Guanjie Zheng, Chang Liu, Hua Wei, Chacha Chen, Zhenhui Li, Rebuilding City-Wide Traffic from Multi-Source Data. In Proceedings of 37th IEEE International Conference on Data Engineering (**ICDE 2021**), Chania, Greece, April 2021.
2. Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD 2020**), Ghent, Belgium, Sep 2020. **(Best Applied Data Science Paper Award)**
3. **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**), New York, United States, February 2020. (Acceptance rate: 1591/7737=20.6%)
4. Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (**CIKM 2019**), Beijing, China, November 2019. (Acceptance rate: 200/1030=19.4%)
5. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2019**), Anchorage, AK, USA, Aug 2019. (Acceptance rate: 170/1200=14.2%)

Workshop Papers

1. **Chacha Chen**, Chieh-Yang Huang, Yaqi Hou, Yang Shi, Enyan Dai, Jiaqi Wang, Joint Event Multi-task Learning for Slot Filling in Noisy Text. In *Proceedings of the Workshop on Noisy User-generated Text* ***(W-NUT, EMNLP 2020)***.
2. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, Deep Reinforcement Learning for Traffic Signal Control along Arterials. In *Proceedings of the Workshop on Deep Reinforcement Learning for Knowledge Discovery* ***(DRL4KDD, KDD 2019)***.

Community Activities & Service

Program Committee

- AAAI 2021: The Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21)
- W-NUT: The 6th Workshop on Noisy User-generated Text with EMNLP 2020

Journal/Conference Reviewer

- AAAI 2020, KDD 2020, CIKM 2020, TITS 2021

Research Experience

IQVIA, Analytics Center of Excellence, Boston

Machine Learning Research Intern — May. 2020–Aug. 2020

- Gaussian Process for health risk prediction: Designed Bayesian machine learning models for informative health risk prediction. Different from previous methods, our proposed model is able to give uncertainty-based rich prediction. Moreover, we leveraged multi-sourced data with different modalities.
- Supervised by Cao (Danica) Xiao (IQVIA), Prof. Fenglong Ma (PSU) and Prof. Jimeng Sun (UIUC).

CityBrain Lab, Hangzhou

Machine Learning Research Intern — Jul. 2018–Aug. 2019

- Reinforcement learning for traffic signal control: Designed and implemented a reinforcement learning (RL) approach with justification on state and reward design for multi-intersection traffic signal control
- Supervised by Prof Zhenhui (Jessie) Li (PSU).

SJTU PAMI Lab, Shanghai
Research Assistant — Sep. 2017–Mar. 2018

- Implementation of SVM with indefinite and semi-definite kernels: Coauthored a software package which accommodates Support Vector Machine (SVM) classification algorithms with indefinite kernels and semi-definite kernels. Moreover, we incorporated Sequential Minimal Optimization (SMO) in the traditional SVM algorithm and Provided various kernel functions, including TAHN, TL1, gaussian, polynomial and linear kernels
- Supervised by Prof Xiaolin Huang (SJTU).

Teaching Experience

Pennsylvania State University
Teaching Assistant — Fall 2020
IST 261 (Java Course): Application Development Design Studio I
Prof. Kenneth Huang

Awards

AAAI-20 VISA Research Scholarship	Feb. 2020
AAAI-20 Student Travel Grant	Feb. 2020
Top 25 in the finalist, *KDD Cup 2020 LDR Competition*	Aug. 2020
Distinguised Bachelor Thesis, *Shanghai Jiao Tong University*	Jun. 2019
Academic Excellence Scholarship, *Shanghai Jiao Tong University*	Sep. 2016

Relevant Skills

Technical: Python, C++, Database, Java, Matlab, R